

Q1) [red blue blue red green grey]

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a)

observation	blue	green	grey	red	pd.getdummies(.)
red	0	0	0	1	
blue	1	0	0	0	
blue	1	0	0	0	
red	0	0	0	1	
green	0	1	0	0	
grey	0	0	1	0	

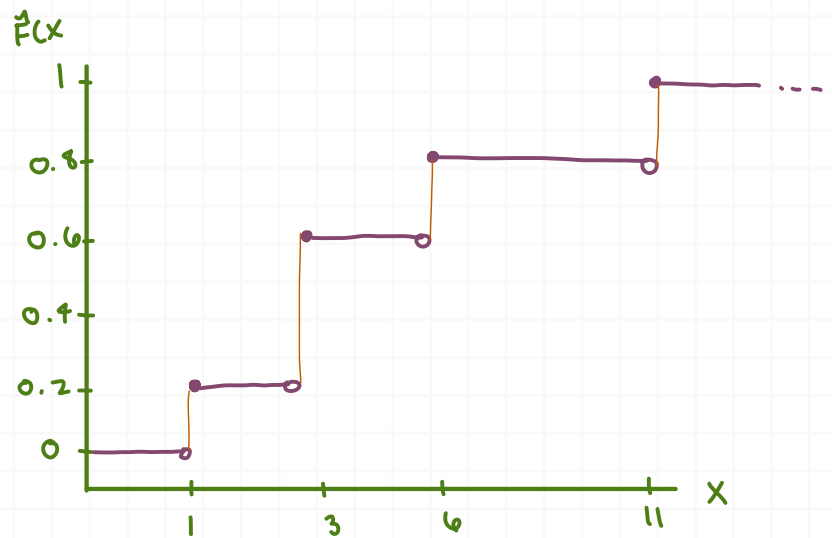
b) sample proportions

label	column sum	proportions
blue	2	$2/6 = 1/3 = .333$
green	1	$1/6 = 0.167$
grey	1	$1/6 = 0.167$
red	2	$2/6 = 1/3 = 0.333$

c) $X = [1, 3, 3, 6, 11]$

	val $\leq X$	ECDF
$X = 1$	1 $\rightarrow$ 1	$1/5 = 0.2$
$X = 3$	1, 3, 3 $\rightarrow$ 3	$3/5 = 0.6$
$X = 6$	1, 3, 3, 6 $\rightarrow$ 4	$4/5 = .8$
$X = 11$	1, 3, 3, 6, 11 $\rightarrow$ 5	$5/5 = 1$

$$\hat{F}(x) = \begin{cases} 0 & \text{if } x < 1 \\ 0.2 & \text{if } 1 \leq x < 3 \\ 0.6 & \text{if } 3 \leq x < 6 \\ 0.8 & \text{if } 6 \leq x < 11 \\ 1 & \text{if } x \geq 11 \end{cases}$$



Q2)

a) $Y = bX + a$

spread/squeezes values,
stretch

shift, slides value
left or right

CDF: $F_Y(y) = P(Y \leq y)$

$F_Y(y) = P(a + bX \leq y)$

$\hookrightarrow bX \leq y - a$

$X \leq \frac{y-a}{b}$

Distr: $F_Y(y) = P(X \leq \frac{y-a}{b}) = F_X(\frac{y-a}{b})$

$\hookrightarrow f_X(\frac{y-a}{b}) \cdot (\frac{1}{b})$

Density: $f_Y(y) = f_X(\frac{y-a}{b}) \cdot (\frac{1}{b})$

$X = [1, 3, 4]$

b) $\frac{1+3+4}{3} = \frac{8}{3} \approx 2.667$

c) $\frac{1+9+16}{3} = \frac{26}{3} \approx 8.667$

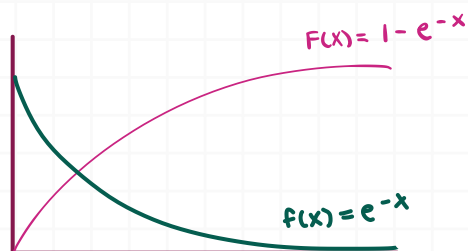
d) $\frac{8^2}{3} = \frac{64}{3} \approx 21.333 \neq 8.667$ b/c averaging gets rid of the extra weight of the large squared values in part b.

$8.667 - 21.333 = -12.666 = \text{variance}$

Q3)

$$F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-x} & x \geq 0 \end{cases}$$

a) density: $f_x(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - e^{-x} & \text{if } x \geq 0 \end{cases}$



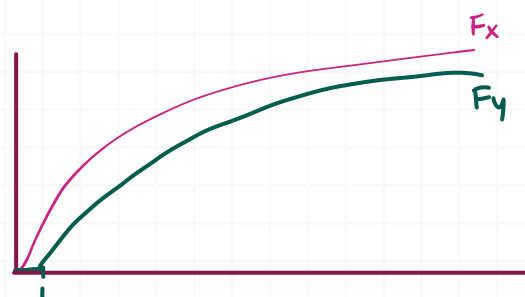
b) $\Pr(X \geq 3) = 1 - \Pr(X < 3) = 1 - F(3) = 1 - (1 - e^{-3}) \approx 0.05$

$\Pr(X \leq 35) = F(35) = 1 - e^{-35} \approx 1$

c) $F_Y(y) = \Pr(1 + 3X \leq y) = \Pr\left(\frac{y-1}{3}\right)$

$$F_Y(y) = F_X\left(\frac{y-1}{3}\right) = \begin{cases} 0 & y < 1 \\ 1 - e^{-(y-1)/3} & y \geq 1 \end{cases}$$

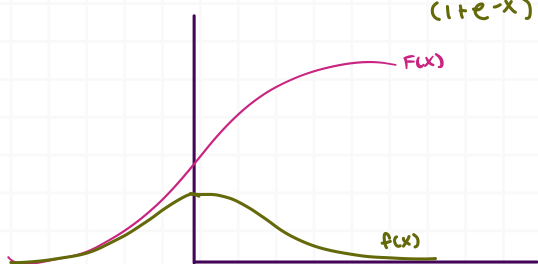
$\Pr(Y < 10) = F_X\left(\frac{10-1}{3}\right) = 1 - e^{-(10-1)/3} \approx 0.95$



Q4)

$$F(x) = \frac{1}{1 + e^{-x}}$$

a) $F(x)' \rightarrow (1 + e^{-x})^{-1} \Rightarrow -(1 + e^{-x})^{-2} \cdot -e^{-x}$
 $\Rightarrow \frac{e^{-x}}{(1 + e^{-x})^2} = f(x)$

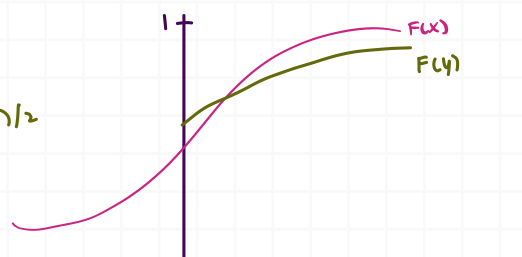


b) $\Pr(X \geq 0.8) = 1 - F(0.8) = 1 - \frac{1}{1 + e^{-0.8}} \approx 0.310$

$\Pr(X \leq 0.3) = F(0.3) = \frac{1}{1 + e^{-0.3}} \approx 0.574$

c) $F_Y(y) = \Pr(2X - 1 \leq X) = \Pr\left(\frac{y+1}{2}\right) \rightarrow F_X\left(\frac{y+1}{2}\right) = F_Y(y) = \frac{1}{1 + e^{-(y+1)/2}}$

$\Pr(Y < 0) = F_Y(0) = \frac{1}{1 + e^{-(0+1)/2}} \approx 0.622$



Q5) a) $1 - e^{-x} = \frac{1}{2}$
 $-e^{-x} = \frac{1}{2} - 1$
 $(-e^{-x} = -\frac{1}{2}) - 1$
 $\ln e^{-x} = \ln \frac{1}{2}$
 $-x = \ln(\frac{1}{2})$
 $x = \ln(2) \approx 0.693$

b) $\frac{1}{1+e^{-x}} = \frac{1}{2}$
 $2 = 1 + e^{-x}$
 $\ln 1 = \ln e^{-x}$
 $\ln(1) = -x = 0$
 $x = 0$

c) $1 - e^{-x} = u$
 $-e^{-x} = u - 1$
 $\ln e^{-x} = \ln(1-u)$
 $-x = \ln(1-u)$
 $F^{-1}(u) = -\ln(1-u)$

d) $\frac{1}{1+e^{-x}} = u$
 $1 + e^{-x} = \frac{1}{u}$
 $e^{-x} = \frac{1}{u} - 1$
 $e^{-x} = \frac{1}{u} - \frac{1}{1} = \frac{1}{u} - \frac{u}{u} = \frac{(1-u)}{u}$
 $\ln e^{-x} = \ln \frac{(1-u)}{u}$
 $-1 \cdot -x = \ln(\frac{1-u}{u}) \cdot -1$
 $x = -\ln(\frac{1-u}{u}) = \ln(\frac{u}{1-u})$
 $F^{-1}(u) = \ln(\frac{u}{1-u})$